Assignment P4

Julia Nething
nething@gatech.edu

1 QUESTION 1

1.1 Initial Situation

The initial situation is the need to contact the professor (in this example, Dr. Joyner) to ask for explanation of a grade. (Note: in many of the methods below, I assume that the user does not remember the email address or Reddit username of staff).

1.2 Selection Rules

- If the class discussion forum, Ed, is currently available and none of the other special circumstances listed below apply, select the "Post on Ed (Discussion Forum)" method.
- If the class discussion forum, Ed, is currently down/unavailable, the message cannot wait, and none of the other special circumstances listed below apply, choose the "Send Email to djoyner3@gatech.edu" method.
- If Dr. Joyner has indicated that he is out of office and will not be monitoring official means of communication, but you insist on getting in contact with him as soon as possible anyway, choose the "Send Direct Message on Reddit to u/DavidAJoyner" method.
- If you feel that Dr. Joyner has committed a violation and you want to also accuse him of this violation officially and have someone contact him on your behalf, choose the "Send Email to OMSCS Associate Director for Student Affairs, Jay Summet, at summetj@gatech.edu" method.

1.3 Methods

1.3.1 *Post on Ed (Discussion Forum)*

Operators:

- Open browser and navigate to edstem.com (1-10 seconds)
- Select the CS6750: Human-Computer Interaction class (1-10 seconds)
- Click the "New Thread" button (1-10 seconds)
- Write out message and subject asking for the explanation of the grade (1-5

minutes)

- Click the "Private" checkbox (1-10 seconds)
- Click the "Post" button (1-10 seconds)

1.3.2 *Send Email to djoyner3@gatech.edu*

Operators:

- Open browser and navigate to Google.com (1-10 seconds)
- Search "david joyner omscs" on Google (1-10 seconds)
- Click on the result that links to Dr. Joyner's page on the Georgia Tech College of Computing website (1-10 seconds)
- Find Dr. Joyner's email (djoyner3@gatech.edu) on that page and copy it to the clipboard (1-10 seconds)
- Navigate in the browser to the email client of choice, e.g. GMail (1-10 seconds)
- Click the "compose" button (1-10 seconds)
- Paste the email address from the clipboard into the "to" section (1-10 seconds)
- Write out message and subject asking for the explanation of the grade (1-5 minutes)
- Click the "send" button (1-10 seconds)

1.3.3 *Send Direct Message on Reddit to u/DavidAJoyner*

Operators:

- Open browser and navigate to Google.com (1-10 seconds)
- Search "david joyner omscs reddit" on Google (1-10 seconds)
- Click on the result that links to Dr. Joyner's Reddit account, u/DavidAJoyner (1-10 seconds)
- Click the "chat" button (1-10 seconds)
- Write out message asking for the explanation of the grade (1-5 minutes)
- Click the "send" button (1-10 seconds)

1.3.4 *Send Email to OMSCS Associate Director for Student Affairs, Jay Summet, at summetj@gatech.edu*

Operators:

- Open browser and navigate to Google.com (1-10 seconds)
- Search "jay summet omscs" on Google (1-10 seconds)
- Click on the result that links to Dr. Summet's page on the Georgia Tech College

of Computing website (1-10 seconds)

- Find and click on the link that takes you to Dr. Summet's personal webpage
- Find Dr. Summet's email (summetj@gatech.edu) on that page and copy it to the clipboard (1-10 seconds)
- Navigate in the browser to the email client of choice, e.g. GMail (1-10 seconds)
- Click the "compose" button (1-10 seconds)
- Paste the email address from the clipboard into the "to" section (1-10 seconds)
- Write out message and subject; this message should outline the violation then ask Dr. Summet to ask Dr. Joyner about your grade on your behalf (1-10 minutes)
- Click the "send" button (1-10 seconds)

1.4 Ultimate Goal

The professor, Dr. Joyner, has been contacted and has received the ask for an explanation of a grade.

2 QUESTION 2

For all tasks/operators below, the time is estimated to be 1-10 seconds unless indicated otherwise.

2.1 Submitting this Assignment to Canvas

1. Submit this assignment to Canvas
    (a) Navigate to the assignment submission page
        i. Navigate to Canvas
            A. Open a browser
            B. Enter the URL (canvas.gatech.edu) into the search bar
            C. Click "search" or hit the "Enter" button
        ii. Log in
            A. Click the "Login to Canvas" link
        iii. Navigate to the submission page
            A. Click on the class's card on the dashboard (CS6750: Human-Computer Interaction)
            B. Click on "Assignments" in the navigation list
            C. Click on "Assignment P4" (the assignment you wish to submit)
            D. Click on "Start Assignment"
    (b) Submit the assignment

i. Upload the assignment file
   A. Click on "Choose File"
   B. Select the correct file
ii. Fill out the rest of the form
   A. Click the checkbox next to the EULA (End-User License Agreement)
   B. Fill out the "comments" text box
   C. Click "Submit Assignment"

2.2 Receiving One's Grade and Feedback

1. Receive one's grade and feedback
   (a) Receive email that grades are on Canvas (up to 4 weeks)
   (b) Navigate to the course's grades page
      i. Navigate to Canvas
         A. Open a browser
         B. Enter the URL (canvas.gatech.edu) into the search bar
         C. Click "search" or hit the "Enter" button
      ii. Log in
         A. Click the "Login to Canvas" link
      iii. Navigate to the grades page
         A. Click on the class's card on the dashboard (CS6750: Human-Computer Interaction)
         B. Click on "Grades" in the navigation list
   (c) Receive the grade and feedback
      i. Read the grade
         A. Scroll down to find the row that says "Assignment P4"
         B. Read the grade in the third column from the left
      ii. Read the feedback
         A. Click on the discussion icon in the fifth column from the left
         B. Read the feedback that pops up in the row below

3 QUESTION 3

3.1 Married Couple and a Map

3.1.1 *Driver*

The driver is reasoning over all of the information given to them and then acting on the car based on that reasoning. For example, they are listening to the passenger's directions while also reading the road signs (such as exit signs). They reason over that information, decide which action to take (e.g. take the exit), and then act on the car to follow that action (e.g. turn on their blinker, move the steering wheel to the right to take the exit). They are also monitoring the dashboard for errors and for how much gas they have left, because this might make them change their goal (for example, their new goal might be to find a gas station).

3.1.2 *Passenger*

The passenger reasons over the map and the road signs to determine where the car is currently located on the map. They then tell the driver the information that they have reasoned so that the driver can act on it accordingly (for example, take the next exit).

3.1.3 *Map*

The map is holding in memory the geographical landscape as a whole, for example all of the exit numbers are their corresponding locations. The passenger does not have to remember that I-10 takes them to the beach and not I-110; they just have to be able to read the map, follow both interstate lines with their eyes, and see which one leads to the beach.

3.1.4 *Road Signs (Exit Signs, etc.)*

The road signs are holding in memory the current location that the car (and driver/passenger) are in. The driver does not have to constantly remember where they are, the exit signs show them things like "1.5 miles to Exit 354" or "Exit 353: Orlando Science Center", which helps the driver understand if this is the exit they want to take or how far the next exit is.

3.1.5 *Car's Dashboard*

The dashboard perceives the current state of the car including how much gas it has and whether or not the blinker is turned on, and then displays that information accordingly. For example, it will show a blinking arrow light point right if the right blinker is turned on. It will also show a dial pointing to how much gas the car has left in its tank. The driver can then use this information and reason over it to determine if they should keep going or stop at the next gas station, for example.

3.2 Lone Driver and a GPS

In both situations, the driver is acting on the world and reasoning over the information that they see (such as how much gas they have in the tank, the road signs, etc.) and what the navigator (either the passenger or the GPS) is telling them. Also, depending on the accuracy of the GPS, map, and the passenger's map reading skills, in both systems the driver is getting relatively good information but not necessarily perfect information (for example, maybe the GPS has not updated their maps for this area so some things are out-of-date, or perhaps the passenger is using an old map).

However, in the married couple example, the passenger can often be wrong. Perhaps they are misreading the map and think that they need to take I-110 instead of I-10. They will tell the driver to exit to I-110, and the driver might believe them because they do not have the map in front of them to double check. A GPS is more accurate in this sense and will compare the car's current location to their destination to give them accurate directions.

On the other hand, a passenger can help the driver make judgment calls that the GPS cannot. For example, if there is construction and it looks like their exit is blocked, a GPS might not know that and tell them to take that exit anyway. A passenger, meanwhile, would be able to see that and be able to quickly reroute them based on the map. In addition, a passenger can visually notice good stopping points and might advise the driver to take the next exit so they can eat at a particular restaurant, which is something that the GPS cannot do. Therefore, the social relationships affect the success of the system as a whole because they can make quick judgment calls based on unexpected obstacles (e.g. construction) or competing goals (e.g. eating lunch).

4 QUESTION 4

4.1 Description of the Task

The task that I will be analyzing for this question is registering for a class on OSCAR. On OSCAR, the user can navigate to different pages where they can search for and filter classes, see the current registration numbers for classes, register for classes, and then see the classes that they are currently registered for for different semesters (e.g. for the summer and the fall). Another piece of this system is OMSCentral, where students write and read reviews of courses.

4.2 Pieces of the System

4.2.1 *The current user*

The current user is the user who is trying to register for a class. In this system, they are reasoning over the information available to them to decide which class to take. They consult multiple pages - the OSCAR page that lists the available courses (which also shows which courses have availability), the user's previous course history, and OMSCentral to decide which course fulfills requirements for them and looks like a wise choice based on reviews. They then act on the system by submitting the form to register for a class.

4.2.2 *Other students*

Other students perceive the amount of time a class takes (in hours/week) and the difficulty level of a class (on a scale of 1-5), and then they write their respective views of these numbers on OMSCentral. They area also registering for classes at the same time, which means that they are also acting on the system and changing the availability of classes at the same time the user is, which may affect if the user is able to register for a class.

4.2.3 *The OSCAR page that lists the available courses*

This OSCAR page lists the available courses, their names and professors, the campus that course is on, the number of students registered, the number of spots available, the number of students waitlisted, and the number of waitlist spots available. They are part of the system's memory because they can display information about what classes are even available and their details. They also perceive the current state of availability for each class.

4.2.4 *The user's account information in OSCAR*

The user's account information holds in memory what classes the user is allowed to take based on the program they are in and what classes they have taken previously.

4.2.5 *The registration page in OSCAR*

This is the page where the user will take a final action by submitting a request to register for the class. It is the page itself though that is acting by making the request to Georgia Tech servers and registering that student in the class, and then updating the enrollment numbers so that other students know if the course is full or not. If the student attempts to register for a class that they are not eligible for, it will reason on the request, comparing the request to a variety of rules, and send back an error.

4.2.6 *OMSCentral*

OMSCentral holds in memory the list of all classes, the amount of time a class takes (in hours/week) and the difficulty level of a class (on a scale of 1-5). Classes can be sorted by these attributes or filtered by the specialization they fulfill, which means they help users reason over which class they want to take.